This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

Single-Point Supervised Oriented Object Detection in Remote Sensing Images via Joint Semantic and Structural Awareness

Yifan Xi, Ting Lu, *Member, IEEE,* Shutao Li, *Fellow, IEEE*

***Abstract*—With the rapid advancement of oriented object detection in remote sensing imagery, weakly supervised learning has shown great promise for reducing the costs of annotation. Among these, point-based supervision is the most efficient. However, the limited positional information with point labels makes it difficult to capture important target attributes, restricting the object detection accuracy. Focused on this, this paper proposes an effective single-point supervised oriented object detection network, where category semantic information and structural edge prior are introduced. First, we design a semantic-guided point assignment strategy based on scarce single-point cues, aiming to drive the network to better predict angle and scale information via matching between instance targets and prior points. Specifically, class-wise semantic information derived from the forward inference process is used to construct a mask for selecting high value prior points, which are then clustered with instance-level point labels as centers to optimize the precise assignment of prior points. Then, we further develop a multi-scale attention mixture-of-experts module to adaptively fuse multi-layer feature maps for learning features of variable scales through a dynamic weighting mechanism. Moreover, a structure-guided Voronoi partition approach is presented, where a structure prior is primarily obtained through gradient computation, and then object shape estimation is refined to converge to the reference outline based on Voronoi partition and segmentation results. Overall, the incorporation of semantic and structural prior knowledge can help to more accurately locate and classify objects with single-point supervision. Extensive experiments on public datasets validate the superiority of the proposed framework. The source code will be available at https://github.com/fan609/S2GPD.**

***Index Terms*—Remote sensing imagery, object detection, weakly supervised learning, point supervision.**

I. Introduction

HIGH precision oriented object detection (OOD) [1], [2], [3], is a key technology for intelligent interpretation of remote sensing images. Accurately locating oriented targets and predicting their posture boosts the application value of remote sensing images in geographic mapping, urban planning, etc. However, achieving precise object detection relies on numerous precise and high-quality annotation bounding boxes. Especially in the complex sceneries with different objects of varying scales and shapes, the annotation process often consumes substantial human and time resources.

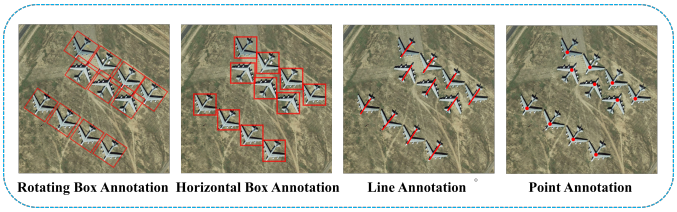


Fig. 1. Different supervision methods for oriented object detection

In order to save annotation costs, weakly supervised oriented object detection technology provides an efficient solution for this, which uses weakly annotated signals that are easier to label (such as horizontal boxes [4], lines [5], and points [6]). As shown in Fig. 1, different supervision signal annotation schemes are presented. The cost of rotated bounding box annotation is 36.5% higher than that of horizontal bounding box annotation and 104.8% higher than that of point annotation [7]. In addition, horizontal bounding box annotation and line annotation are obviously more complex to operate than point annotation. Therefore, single-point supervision is a more efficient and challenging research direction.

Currently, the research field of single-point supervision can be roughly divided into three main directions: 1) Approaches relying on fundamental large models such as SAM (Segment Anything Model) [8]. This approach activates SAM [9] to generate target masks by inputting point prompts, thereby obtaining rotated bounding boxes. However, this method has certain limitations in inference speed and requires a large number of post-processing steps. 2) Strategies assisted by artificial prior knowledge [10]. This strategy integrates manually constructed prior knowledge, for example, developing corresponding templates for different categories, to learn the target scales and orientations indicated by this prior knowledge. Nevertheless, this technique requires designing specific templates for various datasets, indicating it lacks generality and adaptability. 3) Research methods based on intrinsic knowledge attributes of images [11], [12]. This approach guides the model to learn target sizes and orientations by mining intrinsic attributes of images (e.g., structure and layout). In contrast,

This work was supported in part by the National Natural Science Foundation of China under Grant 62271196 and Grant 62221002, and in part by the Outstanding Young Talents Program of Changsha under Grant kq2506006, and in part by the Hunan Provincial Key Laboratory under Grant 2018TP1013, and in part by the Science and Technology Innovation Program of Hunan Province under Grant 2025RC3084. *(Corresponding author: Ting Lu.)*

Yifan Xi, Ting Lu, and Shutao Li are with the College of Artificial Intelligence and Robotics, Hunan University, Changsha 410082, China, and also with the Key Laboratory of Visual Perception and Artificial Intelligence of Hunan Province, Hunan University, Changsha 410082, China (e-mail: ting_lu@hnu.edu.cn).

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

the final approach not only eliminates the need for manual prior knowledge but also avoids complex reasoning steps, thereby exhibiting good generality. Among them, PointOBB [6] exploits image multi-view consistency and multi-instance learning strategies to acquire the orientation and scale of targets. PointOBB-v2 [11] outlines object contours via class probability maps (CPMs). Subsequently, it creates pseudo-rotated boxes from CPMs via weighted sampling and PCA-based (principal component analysis) orientation. Building upon PointOBB, PointOBB-v3 [12] incorporates the prediction branch to enable end-to-end training. Meanwhile, it introduces scale-adaptive feature fusion and instance-aware weighting strategies to alleviate the impact of low-quality pseudo-labels. Point2RBox-v2 [13] adopts a layout-constrained perspective to learn the sizes and orientations of objects in different scenes. However, owing to the inherently limited information content carried by a single point, these methods fail to achieve simultaneous compatibility with extreme scenarios (e.g., extremely sparse object distributions and highly dense ones). Therefore, there is an urgent need to explore an effective approach to enhance the adaptability of the point-supervision method in complex situations.

To address the above challenges, we propose a point-supervised oriented object detection (OOD) framework guided by semantic and structural knowledge (S^2GPD). First, to address the challenge of estimating target scale information and angle information, the multi-scale attention mixture-of-experts module (MS-Attn-MoE) and the semantic-guided point label assignment strategy (SGPA) are designed. Among them, the MS-Attn-MoE utilizes different receptive field convolution kernels to extract variable-scale target features and computes dynamic fusion weights via expert sub-modules for adaptive integration. The SGPA integrates classification heatmap-derived semantic cues to filter high-value prior points, then applies point-guided feature clustering to match prior points and instances accurately. Additionally, the rotation, scaling, and flipping of images enable the network to learn the variability in object angle and scale. Finally, a structure-guided Voronoi partition (SGVP) is presented to improve the object shape perception. It incorporates edge gradient-based structural priors to guide contour approximation and foreground-background separation in clustered regions, while segmentation techniques refine object shapes for precise boundary localization. The effectiveness of this method has been validated on the DOTA-v1.0, DOTA-v1.5, DIOR, and HRSC public datasets.

1) We introduce an end-to-end network for oriented object detection that relies solely on single-point supervision. The network delivers more accurate object predictions by synergistically fusing semantic knowledge with structural priors, which provides a new framework for handling the issue of object detection with limited point label cues.
2) To more precisely estimate varying orientations and scales of targets, we propose the multi-scale attention mixture-of-experts (MS-Attn-MoE) module and the semantic-guided point label assignment (SGPA) strategy. In brief, the MS-Attn-MoE dynamically fuses multi-scale features via an adaptive weighting mechanism, and the SGPA incorporates classification semantic information to match between high-value prior points and instance targets.
3) For further enhancing the perception capability of the target-specific shape characters, a structure-guided Voronoi partition (SGVP) method is designed. The SGVP leverages the gradient-based structural information to guide Voronoi partition, yielding the approximate scope of targets, and subsequently precisely delineates object contours.

The remainder of this paper is organized as follows: Section II reviews and analyzes relevant works. In Section IV, we present the details of the proposed method. Section IV conducts detailed experiments and analyzes the results. Finally, conclusions are drawn in Section V.

II. Related Works

A. Fully-Supervised Oriented Object Detection

Oriented object detection has emerged as a critical research focus following advances in horizontal bounding box detection. Current methodologies can be categorized into four primary paradigms: two-stage frameworks [14]–[16], anchor box-based one-stage detectors [17]–[19], anchor-free one-stage approaches [20]–[22], and Transformer-based architectures [23]–[25]. The two-stage framework in object detection is an established paradigm that divides the process into two sequential steps, namely region proposal generation followed by refined object classification and regression. The RoI Transformer was introduced by Ding *et al.* [26] to derive rotated regions of interest (RoIs) from horizontal proposals, incorporating a dedicated rotated RoI alignment module (RRoI Align) for oriented feature extraction. Xie *et al.* [27] formulated Oriented R-CNN, which generates rotated RoIs with minimal computational overhead and utilizes an R-CNN detection head for high-precision localization. Anchor-based one-stage detectors are end-to-end detection architectures that directly perform simultaneous object localization and classification on feature maps via pre-defined multi-scale anchor boxes. To mitigate rotation-sensitive errors and loss discontinuity in five/eight-parameter systems, Qian *et al.* [28] presented a rotation-sensitive detection network (RSDet) with modulated loss and an eight-parameter single-stage architecture. Yang *et al.* [29] proposed a feature refinement module for one-stage alignment and an approximate Skew IoU loss to address non-differentiability in oriented IoU computation. Anchor-free one-stage methods are a class of single-stage detection schemes that achieve object detection directly through keypoint prediction or center region regression without pre-defined anchor boxes. For example, Yao *et al.* [30] constructed a hierarchical mask prompting mechanism for instance-background separation, employing oriented IoU loss to resolve angle insensitivity in traditional IoU metrics. Li *et al.* [31] leveraged adaptive point sets to encode instance contours, circumventing explicit angle prediction. Liang *et al.* [32] introduced an compact anchor-and-angle-free detector, MidNet, based on a midpoint representation, that uses symmetric deformable convolutions to

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

effectively eliminate unstable angular regression and the need for excessive anchor settings. The Transformer-based object detection framework fundamentally reformulates detection as an encoder-decoder sequence prediction task, leveraging self-attention mechanisms to model global contextual dependencies while eliminating hand-designed components through end-to-end set prediction. For instance, Zhu *et al.* [33] enhanced DETR's perception of oriented objects via rotation-sensitive deformable attention in feature alignment and decoder components. Lee *et al.* [34] devised a Hausdorff-distance-based cost function for DETR's Hungarian matching, improving geometric discrepancy measurement between predictions and ground truth. Although these methods improve object detection performance by upgrading model architectures, they often rely on substantial amounts of high-quality supervised information.

B. Weakly-Supervised Oriented Object Detection

In the current field of weakly supervised oriented object detection, there exist two primary paradigms: horizontal bounding box supervision, line supervision, and point supervision. The horizontal box supervision framework implements oriented object detection using horizontal boxes as supervision signals. Among them, Yang *et al.* [35] constrained candidate angles using the geometric symmetry of objects and subsequently filtered precise angles by introducing a self-supervised branch. Yu *et al.* [36] introduced a self-supervised branch based on flip and rotation consistency to learn angles by leveraging visual object symmetry. They further combined an improved weakly supervised branch and the CircumIoU loss to enhance oriented object detection performance under horizontal bounding box supervision. Wang *et al.* [37] proposed the EIE-Det method, which achieves efficient weakly supervised oriented object detection using only horizontal bounding box annotations through explicit and implicit box equivariance learning and Tanimoto coefficient regression loss, with performance approaching that of fully supervised models. The line-based supervision framework utilizes oriented line segments covering target objects as supervision signals to achieve oriented object detection. For example, Fu *et al.* [5] proposed the first line-supervised oriented object detector L2RBox with an end-to-end anchor-free architecture. They further designed line annotation-adapted label assignment (mapping lines to circle domains for training sample selection and centerness target calculation) and a regression loss. The point-based supervision framework realizes oriented object detection by using a central point or any arbitrary point within the target object as the supervision signal. For instance, both the P2RBox proposed by Cao *et al.* [38] and PMHO proposed by Zhang *et al.* [8] leveraged the strong zero-shot generalization capability of the SAM in point-to-mask tasks to achieve oriented object detection. Luo *et al.* [6] achieved the process of point-to-box generation by leveraging multi-view scale consistency and multi-instance learning (MIL) strategies. Yu *et al.* [10] adopted a knowledge fusion strategy, where knowledge related to target size and rotation is learned through sketches, while information relevant to target classification is learned using point annotation data, thereby achieving the generation of rotated boxes. Ren *et al.* [11] determined the orientation and boundary limits of the target by employing class probability heatmaps and principal component analysis (PCA) methods. Yu *et al.* [13] focused on the perspective of target layout constraints, achieving high-precision detection of dense scene objects using point annotations by constructing Gaussian overlap loss, Voronoi loss, and edge loss functions. In the remote sensing video domain, Feng *et al.* [39] proposed the remarkable SDANet for moving-vehicle detection in satellite videos, which suppresses false alarms using weakly supervised road priors and enables adaptive detection through density matching, achieving outstanding performance on dense small-object scenes. Furthermore, Feng *et al.* [40] presented a novel coarse-to-fine dynamic refinement model for point-supervised vehicle detection, which delivers superior detection performance while drastically reducing annotation costs via spatio-guided pseudo-box generation and iterative refinement. The point supervision approach enables annotation through the configuration of a single annotation point, offering greater simplicity and efficiency compared to other weakly supervised signals. However, this supervision method faces greater challenges in terms of prediction granularity.

III. Proposed Approach

As illustrated in Fig. 2, this is the workflow of the S^2GPD framework. First, we input both original and augmented view images, where the augmentations include rotation, flipping, and scaling. The rotation and flipping perspectives can be used to learn the angular information of the target. Then, features are extracted using the backbone network and FPN (Feature Pyramid Networks), subsequently the three-scale features are input to the MS-Attn-MoE module. This module not only facilitates cross-scale feature fusion but also enhances the capability of single-scale feature maps to perceive target features of varying scales. To further improve object detection performance, the SGPA strategy is proposed, which leverages semantic information from classification heatmaps to guide the assignment of high-value prior points and provides valuable semantic cues for estimating target scale and angle. Finally, the SGVP method is introduced, which integrates structural prior information into Voronoi partitioning to strengthen the model's ability to perceive target shapes. Moreover, the Voronoi loss and edge loss [13] are adopted to enhance object scale perception. The angular estimation of the target is optimized based on the multi-view consistency loss [13].

A. Semantic-Guided Point Assignment (SGPA)

To mitigate the significant deviation in target scale and angle under point-based supervision prediction in both sparse and crowded scenes, a point label assignment strategy based on classification semantic guidance is proposed. It dynamically selects prior points with high semantic value as candidates through heatmap activation values. This enables the label assignment process to integrate both spatial location information and semantic structure information, thereby enhancing the model's perception capability and prediction accuracy

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

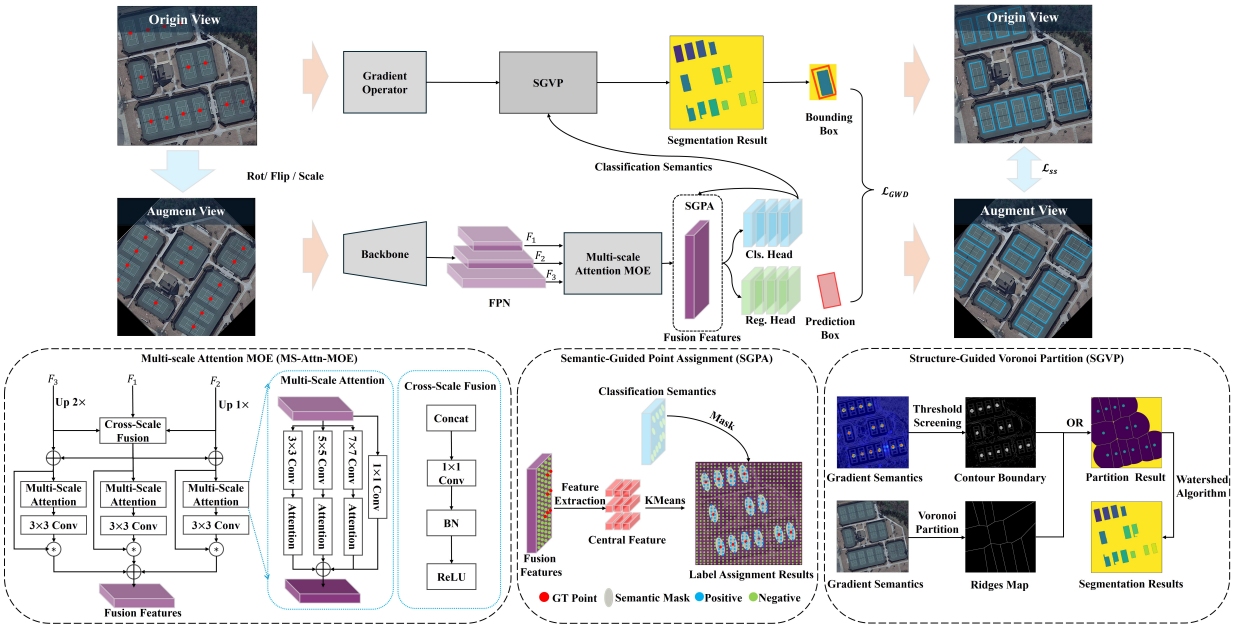


Fig. 2. The proposed semantic and structural knowledge-guided framework for oriented object detection under point supervision.

for target attributes. The detailed optimization approach is outlined below.

First, the coordinates of the prior points on the feature map are denoted as $\mathbf{P} \in \mathbb{R}^{N\times 2}$. The center points of the rotated targets are denoted as $\mathbf{G} \in \mathbb{R}^{M\times 2}$. N and M denote the number of prior points and the number of annotated target center points, respectively. The dimension alignment on $\mathbf{P}$ and $\mathbf{G}$ is performed, expanding their dimensions to $N \times M \times 2$. The offsets $\mathbf{O}$ from each prior point to the target center points are calculated as below:

$$\mathbf{O} = \mathbf{P} - \mathbf{G} \tag{1}$$

The predicted bounding boxes obtained from model inference on annotated images are treated as pseudo ground truth (GT) boxes. These GT boxes are denoted as $\mathbf{Box}_{GT}$ and similarly expand their dimensions to $N \times M \times 5$. The range of bounding boxes $\mathbf{Box}_{prior}$ ($\mathbf{l}$, $\mathbf{r}$, $\mathbf{t}$, $\mathbf{b}$) for ground truth (GT) boxes is computed in a coordinate system centered at the prior points.

$$\begin{aligned} \mathbf{l} &= \frac{\mathbf{w}}{2} + \mathbf{O}_x, \quad \mathbf{r} = \frac{\mathbf{w}}{2} - \mathbf{O}_x \\ \mathbf{t} &= \frac{\mathbf{h}}{2} + \mathbf{O}_y, \quad \mathbf{b} = \frac{\mathbf{h}}{2} - \mathbf{O}_y \end{aligned} \tag{2}$$

where $\mathbf{O}_x$ and $\mathbf{O}_y$ denote the offsets $\mathbf{O}$ along the x-axis and y-axis, respectively. Based on the predefined central sampling region and the current level of the feature map, the sampling radius is set as $s \times r$, where s represents the scaling factor between the feature map at the current level and the resolution of the original image, and r is a predefined value within the specified sampling radius range [0, 1]. Subsequently, by setting the sampling radius, we obtain the mask $\mathbf{M}_{gt}$ of positive sample points located within the GT boxes.

$$\mathbf{M}_{gt} = \mathbb{I}\{\mathbf{O} < s \times r\} \tag{3}$$

where the symbol $\mathbb{I}\{\cdot\}$ denotes the indicator function, a function that returns 1 when the condition within parentheses is satisfied, and 0 otherwise. The symbol explanation also applies to the subsequent formula. Similarly, according to the predefined regression range $[T_1, T_2]$ of the current layer, a mask $\mathbf{M}_r$ for bounding boxes satisfying the condition is obtained.

$$\mathbf{M}_r = \mathbb{I}\left\{T_1 < \mathbf{Box}_{prior} < T_2\right\} \tag{4}$$

Then, a classification heatmap $\mathbf{F} \in \mathbb{R}^{B\times C\times H\times W}$ is introduced as semantic guidance. Here, B denotes the batch size, C represents the number of classes to be classified, and H and W are the dimensions of the classification feature map. The corresponding classification maps $\mathbf{F}_i \in \mathbb{R}^{B\times H\times W}$ are extracted according to the categories of the annotated points, where i represents the index of the category that the current annotation point belongs to. To obtain the main activation regions of $\mathbf{F}_i$, we first apply the activation function to $\mathbf{F}_i$ and then leverage its statistical properties to derive the foreground activation regions.

$$\widehat{\mathbf{F}}_i = Sigmoid(\mathbf{F}_i) \tag{5}$$

$$\mathbf{M}_{fg} = \mathbb{I}\left\{\widehat{\mathbf{F}}_i > mean(\mathbf{F}_i) + std(\mathbf{F}_i)\right\} \tag{6}$$

where $mean\,(\cdot)$ and $std\,(\cdot)$ denote the operations of computing the mean and standard deviation, respectively. $\mathbf{M}_{fg}$ represents the foreground region mask of the classification heatmap. Following this, the clustering algorithm is employed to cluster the features $\mathbf{F}_{MoE}$ output by MS-Attn-MoE. It is only applied when the $\widehat{\mathbf{F}}_i$ contains multiple instances. For $\widehat{\mathbf{F}}_i$ with a single instance, the label is assigned based on the activated region of the mask $\mathbf{M}_{fg}$. The algorithm of points-guided feature clustering is illustrated in **Algorithm** 1. Within this clustering procedure, the foreground object feature is extracted using the mask $\mathbf{M}_{fg}$, and its mathematical formulation is expressed as follows:

$$\mathbf{F}_{fg} = \mathbf{F}_{MoE} \odot \mathbf{M}_{fg} \tag{7}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

where $\mathbf{F}_{fg}$ denotes the foreground point target feature, which is also extracted from $\mathbf{F}_{MoE}$. The symbol $\odot$ indicates element-wise multiplication. And the explanation of this operation also applies to the subsequent formula. Clustering is performed on foreground regions of the classification heatmap, followed by label assignment for each instance, according to the order of real annotated points.

Algorithm 1 Points-Guided Feature Clustering

Input:

$\mathbf{F}_{fg} \in \mathbb{R}^{N_{fg} \times D}$ denotes the foreground target features from $\mathbf{F}_{MoE}$.
$\mathbf{F}_{gt} \in \mathbb{R}^{K \times D}$ represents the features of the target annotation points.
K denotes the number of annotation points.
T is the maximum number of iterations.

Output:

$\mathbf{C} \in \mathbb{R}^{K \times D}$ is the cluster center features.
$\mathbf{l} \in \mathbb{R}^{N_{fg} \times 1}$ is the clustering labels of target points.

if $\mathbf{F}_{gt} = \emptyset$ **then**
  $\mathbf{S} \leftarrow$ Generate a random permutation ranging from 0 to $N_{fg} - 1$.
  $\mathbf{C} \leftarrow$ Filter the first K elements from $\mathbf{F}_{fg}$ according to sequence $\mathbf{S}$ as the initial features.
else
  $\mathbf{C} = \mathbf{F}_{gt}$
end if
while $t \leqslant T$ **do**
  $\mathbf{F}_{fg} \leftarrow$ Expand the dimension of $\mathbf{F}_{fg}$ to $N_{fg} \times K \times D$.
  $\mathbf{F}_{gt} \leftarrow$ Expand the dimension of $\mathbf{F}_{gt}$ to $N_{fg} \times K \times D$.
  $\mathbf{D}_{cos} \leftarrow$ Calculate the cosine similarity of $\mathbf{F}_{fg}$ and $\mathbf{F}_{gt}$ along the second dimension.
  $\mathbf{D}_{dist} = 1 - \mathbf{D}_{cos}$
  $\mathbf{l} \leftarrow$ Record the index of the minimum value in the second dimension of matrix $\mathbf{D}_{dist}(N_{fg} \times K)$
  $\mathbf{C}_{new} \leftarrow$ Calculate new cluster centers based on labels $\mathbf{l}$ and $\mathbf{F}_{fg}$.
  if Sum($\mathbf{C}_{new}$ - $\mathbf{F}_{gt}$) < 1e-7 **then**
    Break
  end if
  $\mathbf{C} = \mathbf{C}_{new}$
end while
return $\mathbf{C}$, $\mathbf{l}$

Finally, we compute the maximum value along the class dimension of the classification heatmap $\mathbf{F}$ to obtain the feature $\mathbf{F}_{max}$ with the shape $B \times H \times W$. Next, the regions where $\mathbf{F}_{max}$ exceeds $\tau_s = 0.65$ are filtered.

$$\mathbf{M}_{max} = \mathbb{I}\{\mathbf{F}_{max} > \tau_s\} \tag{8}$$

The area of the bounding box computed based on the $\mathbf{Box}_{GT}$ is denoted as $\mathbf{A}$ $(N \times M)$. The area outside the regression range $\mathbf{M}_r$ and the GT box range $\mathbf{M}_{gt}$ is set to infinity. As shown in the following equation,

$$\mathbf{A} \odot \mathbb{I}\left\{\mathbf{M}_{gt} = 0\right\} = INF, \quad \mathbf{A} \odot \mathbb{I}\{\mathbf{M}_r = 0\} = INF \tag{9}$$

where INF denotes infinity. Prior points are assigned to the GT box with the smaller area via area $\mathbf{A}$, and the corresponding category is recorded in $\mathbf{L}$.

$$\mathbf{L} = Index(\mathbf{A}.min()), \quad \mathbf{L} \odot \mathbb{I}\{\mathbf{A}.min() = INF\} = C \tag{10}$$

where $min(\cdot)$ takes the minimum value in a given dimension. $Index$ denotes the operation of extracting the subscript of the corresponding element, which here refers to the subscript of the GT box with the smallest area matched by prior points. C refers to assigning the background category to labels where the minimum area is infinity. The label assignment results $\mathbf{l} \odot \mathbb{I}\{\mathbf{M}_{max}\}$ on $\mathbf{l}$ are extracted using mask $\mathbf{M}_{max}$. And we assign the results of the assignment to the corresponding positions in $\mathbf{L}$. Subsequently, semantic information is supplemented via mask $\mathbf{M}_{max}$ to refine the assignment between prior points and labels.

B. Multi-Scale Attention Mixture-of-Experts (MS-Attn-MoE)

Given the challenge of effectively allocating feature maps with the optimal scale for point-annotated targets within FPN layers, most algorithms select the highest-resolution feature map from FPN layers for prediction. Owing to its detailed information, this high-resolution feature map demonstrates superior performance in small target detection. However, its perceptual capability is relatively weaker when extracting large variable-scale target features (e.g., bridges). In response to this, this study proposes an MS-Attn-MOE, which can adaptively fuse features from FPN layers to generate a feature map that integrates both detailed information and global semantics.

First, the two low-resolution feature maps $\widetilde{\mathbf{F}}_1$, $\widetilde{\mathbf{F}}_2$ from the FPN layers are upsampled to align their spatial dimensions with the high-resolution feature map $\widetilde{\mathbf{F}}_3$.

Then, after aligning the feature maps, the cross-scale fusion module is performed across all features. The fused information is respectively added to each layer of feature maps, followed by the implementation of multi-scale attention operations on each feature map. As illustrated in Fig. 2, the multi-scale attention module is primarily composed of three parallel convolutional layers with kernel sizes of 3×3, 5×5, and 7×7, respectively, designed to perceive targets at different scales across feature map levels. To further enhance target saliency, a general-purpose attention module is employed. The attention module is flexibly designed, and in this study, a common channel attention mechanism is adopted. In the final stage, a 1×1 convolutional layer is utilized to implement residual connections, which strengthens feature semantics.

Finally, the feature maps are fused through an MoE. The output feature $\widetilde{\mathbf{F}}_j$ $(1\times h\times w)$ of the multi-scale attention module is fed into a 3×3 convolution (with single-channel output) followed by Softmax activation to produce $\mathbf{G}_j$ $(1 \times h \times w)$, thereby obtaining fusion weights for different scales at the same spatial position. j is the number of layers in the FPN.

$$\mathbf{G}_j = Softmax\left(conv\left(\widetilde{\mathbf{F}_j}\right)\right) \tag{11}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

where $conv$ denotes a 3×3 convolution. The final single-layer feature map $\mathbf{F}_{MoE}$ can be obtained by:

$$\mathbf{F}_{MoE} = \sum_{j=1}^{3} \mathbf{G}_j \cdot \widetilde{\mathbf{F}}_j \tag{12}$$

C. Structure-Guided Voronoi Partition (SGVP)

In order to achieve a more precise estimation of the target contour and provide scale supervision information for training point-supervised models, we propose a Voronoi partition method based on structure prior information. It primarily extracts the contours of the target using edge extraction operators, followed by thresholding to select the more prominent external boundaries while suppressing the relatively weak semantics. This approach helps to increase the distinction between foreground and background recognition. The detailed workflow is described as follows.

To begin with, the target is converted into an anisotropic Gaussian distribution based on its predicted mean and variance. Assume that the predicted target mean is denoted as μ $(n\times2)$ and the covariance as $\mathbf{\Sigma}$ $(n\times2\times2)$. n represents the number of predicted targets. The transformation process is illustrated as follows.

$$\mathbf{E},\mathbf{V} = Eigh(\mathbf{\Sigma}), \quad \mathbf{E}_{norm} = \frac{\mathbf{E}}{\sqrt{\mathbf{e}_1\times\mathbf{e}_2}}\sigma \tag{13}$$

where $Eigh(\cdot)$ denotes the eigendecomposition. $\mathbf{E}$ $(n\times2)$ and $\mathbf{V}$ $(n\times2\times2)$ represent the eigenvalues and eigenvectors, respectively. $\mathbf{e}_1$ and $\mathbf{e}_2$ are the individual eigenvalue elements of $\mathbf{E}$. σ is introduced to preserve the proportionality of eigenvalues after normalization. The covariance matrix $\mathbf{\Sigma}_i^R$ (2×2) is reconstructed as shown in the following equation.

$$\mathbf{\Sigma}_i^r = \frac{\mathbf{V}_i Diag(\mathbf{E}_i)\mathbf{V}_i^T}{D^2} \tag{14}$$

where i denotes the ith target. $\mathbf{E}_i$ and $\mathbf{V}_i$ represent the eigenvalue and eigenvector of the ith target, respectively. $Diag(\cdot)$ means to diagonalize an element. D is the downsampling factor. Based on the mean μ_i and covariance matrix $\mathbf{\Sigma}_i^r$, we transform the ith target into a Gaussian probability distribution.

$$\mathbf{Vor}_i\left(\mathbf{x};\mu_i,\mathbf{\Sigma}_i^r\right) = \frac{1}{2\pi\sqrt{\left|\mathbf{\Sigma}_i^r\right|}}\exp\left(-\frac{(\mathbf{x}-\mu_i)^T(\mathbf{x}-\mu_i)}{2\mathbf{\Sigma}_i^r}\right) \tag{15}$$

where exp is the exponential function. After n targets are processed by function $\mathbf{Vor}_i(\cdot)$, and n Gaussian probability distribution maps $\mathbf{Vor}(n\times h\times w)$ are obtained.

Then, by taking the maximum value of **Vor** along the first dimension, we obtain the maximum value **val** $(1\times h\times w)$ and the corresponding coordinate **ind** $(1\times h\times w)$.

$$\mathbf{val} = max(\mathbf{Vor}), \mathbf{ind}(i_h,i_w) = argmax_{j=1}^{n}\mathbf{Vor}(j,i_h,i_w) \tag{16}$$

where $max(\cdot)$ denotes the maximum value, and $argmax(\cdot)$ denotes the minimum subscript corresponding to the spatial position (i_h,i_w) in the first dimension j of **Vor**. Next, upsampling and expansion operations are performed on **val**, **ind**, and $\widehat{\mathbf{F}}$ respectively to align their spatial dimensions to $H\times W$. The label map **cls** $(H\times W)$ is obtained based on the ground-truth label information of instance points and the instance assignment indices **ind**. To obtain the ridges of the Voronoi diagram, i.e., the target polygon boundaries, the Laplacian operator is employed to extract Voronoi edges, resulting in the ridge map **R**. For the watershed algorithm to accurately perceive the edges of the target, we introduce edge semantic information to supplement the ridge **R**. Here, the $Scharr$ edge operator is employed to extract gradient information **Grad** from the classification heatmap $\widehat{\mathbf{F}}$. Then, the maximum value is taken along the class dimension of **Grad** $(C\times H\times W)$ to obtain $\mathbf{Grad}_{max}$ $(H\times W)$.

$$\mathbf{Grad} = Scharr\left(\widehat{\mathbf{F}}\right), \quad \mathbf{Grad}_{max} = max(\mathbf{Grad}) \tag{17}$$

Next, the gradients are normalized and subjected to a logical OR operation with the ridge map **R** to achieve boundary semantic supplementation.

$$\mathbf{Grad}_{norm} = \frac{\mathbf{Grad}_{max} - min(\mathbf{Grad}_{max})}{max(\mathbf{Grad}_{max}) - min(\mathbf{Grad}_{max})} \tag{18}$$

$$\mathbf{R}_b = \mathbf{R} \vee (\mathbf{Grad}_{norm} > \tau_b) \tag{19}$$

where $\mathbf{R}_b$ denotes the new ridge map after boundary semantic guidance. $\vee$ denotes the binary OR operation. τ_b is the screening threshold, with its value set to 0.7.

Finally, we partition **ind** by setting positive **Pos** and negative thresholds **Neg** for different classes. **Pos** identifies the core foreground region for each instance, where pixels with probabilities above this threshold serve as foreground seed markers. **Neg** identifies definite background regions, where pixels with probability values below this threshold serve as background seed markers. The area between the two thresholds remains unlabeled as a boundary transition zone for the watershed algorithm to refine. To distinguish it from the area to be partitioned, the **ind** is incremented by 1. When processing **ind**, we set to zero the portions where the **val** is below the positive sample threshold **Pos** for the corresponding class **cls**. These regions constitute the predefined areas to be segmented. Conversely, the regions in **ind** where the **val** falls below the negative threshold **Neg** for the corresponding class **cls** are classified as background areas. Simultaneously, the corresponding sections in the new ridge map $\mathbf{R}_b$ are also designated as background regions.

$$\begin{cases} \mathbf{ind}\odot\mathbb{I}\{\mathbf{val} < \mathbf{Pos}[\mathbf{cls}]\} = 0 \\ \mathbf{ind}\odot\mathbb{I}\{\mathbf{val} < \mathbf{Neg}[\mathbf{cls}]\} = n+1 \\ \mathbf{ind}\odot\mathbb{I}\{\mathbf{R}_b\} = n+1 \end{cases} \tag{20}$$

where regions with a value of 0 in **ind** denote the regions to be segmented, and those with a value of $n+1$ represent the background regions. $[\cdot]$ denotes a value retrieval operation based on the number specified in parentheses. The segmentation result is eventually obtained by applying the watershed algorithm to the input image and the labeled image.

$$\mathbf{S} = Watershed(\mathbf{I},\mathbf{G},\mathbf{ind}) \tag{21}$$

where $\mathbf{I}$ is the input image. $\mathbf{G}$ denotes the GT point. $Watershed(\cdot)$ is the watershed algorithm. $\mathbf{S}$ is the output target outline (pixel coordinates) corresponding to each annotated instance. Next, by performing scale transformation on

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

the segmentation points **S** of each target, the target covariance matrix is constructed, and the GWD (Gaussian Wasserstein Distance) [41] loss is calculated with the predicted covariance matrix to achieve accurate scale estimation.

D. Total Loss

The overall training loss consists of the classification loss $\mathcal{L}_{cls}$, the box loss $\mathcal{L}_{box}$ (mainly used for the pseudo boxes of pasted objects in copy-paste augmentation), the Gaussian overlap loss $\mathcal{L}_O$, the Voronoi loss $\mathcal{L}_W$, the edge loss $\mathcal{L}_E$, and the symmetry consistency loss $\mathcal{L}_{ss}$. The total loss is formulated as follows:

$$\begin{aligned}\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \omega_{\text{box}}\mathcal{L}_{\text{box}} + \omega_O\mathcal{L}_O \\ + \omega_W\mathcal{L}_W + \omega_E\mathcal{L}_E + \omega_{\text{ss}}\mathcal{L}_{\text{ss}}\end{aligned} \tag{22}$$

Among them, the Gaussian overlap loss $\mathcal{L}_O$ provides an upper bound on the target size. In densely distributed object regions, the 2D Gaussian distributions of two overlapping objects can be expressed as $\mathcal{N}_1(\mu_1, \mathbf{\Sigma}_1)$ and $\mathcal{N}_2(\mu_2, \mathbf{\Sigma}_2)$. Since objects in remote sensing images rarely overlap, we optimize the overlapping regions of oriented objects to reduce box overlap. The Gaussian overlap loss is formulated as follows:

$$\mathcal{L}_O = \frac{1}{N}\sum_{i\neq j}\exp\left(-\frac{1}{8}\mu^T\mathbf{\Sigma}^{-1}\mu\right)\cdot\frac{|\mathbf{\Sigma}_1|^{1/4}\,|\mathbf{\Sigma}_2|^{1/4}}{\mathbf{\Sigma}^{1/2}} \tag{23}$$

The Voronoi loss $\mathcal{L}_W$ constrains the lower bound of the target size and is defined as:

$$\mathcal{L}_W = \mathcal{L}_{GWD}\left(\begin{bmatrix} w/2 & 0 \\ 0 & h/2 \end{bmatrix}^2, \begin{bmatrix} w_t/2 & 0 \\ 0 & h_t/2 \end{bmatrix}^2\right) \tag{24}$$

where (w, h) is the predicted target size, and (w_t, h_t) is the coarse size derived from watershed segmentation. The GWD loss $\mathcal{L}_{GWD}$ denotes the Gaussian Wasserstein distance. It penalizes the deviation between the predicted box covariance and the covariance derived from the instance region obtained by Voronoi-watershed segmentation.

$$\mathcal{L}_{GWD} = \mathbf{Tr}\left(\mathbf{\Sigma} + \mathbf{\Sigma}_t - 2\left(\mathbf{\Sigma}^{1/2}\mathbf{\Sigma}_t\mathbf{\Sigma}_t^{1/2}\right)^{1/2}\right) \tag{25}$$

Here, $\mathbf{\Sigma}$ and $\mathbf{\Sigma}_t$ denote the predicted target scale variance and the coarsely estimated target scale variance, respectively.

The edge loss $\mathcal{L}_E$ further refines the size estimation capability and is defined as:

$$\mathcal{L}_E = Smooth_{L1}\left(\begin{bmatrix} w & h \end{bmatrix}, \begin{bmatrix} w_e & h_e \end{bmatrix}\right) \tag{26}$$

Here, w_e and h_e are the width and height regression targets, obtained by weighting the edge feature distribution and the Gaussian kernel distribution and selecting the maximum.

The symmetry consistency loss $\mathcal{L}_{ss}$ enforces invariance to target size and orientation by leveraging different augmented views.

$$\mathcal{L}_{ss} = \mathcal{L}_{GWD}\left(\alpha\Sigma\alpha^T, \Sigma_{aug}\right) + \mathcal{L}_{ANG}\left(m\theta + r, \theta_{aug}\right) \tag{27}$$

where $\mathcal{L}_{GWD}$ is Gaussian Wasserstein distance loss, and α is the augmentation transform. Σ and Σ_{aug} are covariance matrices, while $\mathcal{L}_{ANG}$ is angle consistency loss with (m, r) as coefficients and (θ, θ_{aug}) as predicted angles. When rotation is selected, $m = 1$. For flipping, $(m, r) = (-1, 0)$. For scaling, $(m, r) = (1, 0)$.

$$\mathcal{L}_{ANG}(\theta_1, \theta_2) = \min_{k\in\mathbb{Z}}\left(smooth_{L_1}(\theta_1, k\pi + \theta_2)\right) \tag{28}$$

The $\min(\cdot)$ operation avoids the angular periodicity problem.

Following the hyperparameter settings of Point2Rbox-v2, the weights $(\omega_{box}, \omega_O, \omega_W, \omega_E, \omega_{ss})$ are set to $(1, 10, 5, 0.3, 1)$.

IV. Experiments

In this section, we conduct empirical evaluations on the proposed point-supervised oriented object detection algorithm guided by semantic and structural knowledge. Initially, this paper briefly elaborates on the datasets employed in the experiments. Subsequently, the parameter configurations in the experiments across various datasets are described in detail. Finally, through ablation studies and comparative experiments, the advantages of the proposed method are validated and analyzed.

A. Datasets

Here, the four publicly available oriented object detection datasets include DOTA-v1.0 [42], DOTA-v1.5 [42], DIOR-R [43], and HRSC [44]. We will briefly describe these datasets below.

HRSC: The HRSC2016 dataset, developed by Northwestern Polytechnical University, is a benchmark dataset for rotated ship detection in high-resolution aerial imagery. It comprises 1,061 images with resolutions ranging from 300×300 to 1500×900 pixels, containing 2976 annotated ship instances captured under diverse maritime conditions such as varying lighting and harbor clutter. Unlike conventional datasets, HRSC2016 uses oriented bounding boxes (OBBs) to encode ship positions, rotation angles, and bow coordinates, addressing the geometric complexity of elongated maritime targets. The dataset is split into training (436 images), validation (181 images), and test (444 images) subsets, enabling robust evaluation of angle-sensitive detection models.

DOTA-v1.0: The DOTA-v1.0 is a comprehensive remote sensing object detection dataset designed to evaluate algorithms under complex scenarios with diverse object scales and orientations. It includes RGB and grayscale images with resolutions from 800×800 to 20000×20000 pixels, featuring 188282 annotated instances across 15 categories. The 15 categories include plane (PL), baseball diamond (BD), bridge (BR), ground track field (GTF), small vehicle (SV), large vehicle (LV), ship (SH), tennis court (TC), basketball court (BC), storage tank (ST), soccer ball field (SBF), roundabout (RA), harbor (HA), swimming pool (SP) and helicopter (HC). Each instance is labeled using clockwise-ordered quadrangle bounding boxes, supporting both horizontal and oriented bounding box detection tasks. For training and evaluation, DOTA-v1.0 provides single-scale and multi-scale splits: the single-scale version includes 21046 training, 5297 validation, and 10833 test images, while the multi-scale version comprises 138883 training, 35462 validation, and 71888 test images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

DOTA-v1.5: The DOTA-v1.5 extends DOTA-v1.0 as an enhanced remote sensing dataset optimized for small target detection. It introduces supplementary annotations for objects smaller than 10 pixels, refines small-object labeling, and expands the category set to 16 by adding container crane (CC). The total instance count is increased to 403318, strengthening its role as a comprehensive benchmark for evaluating small-target detection algorithms. DOTA-v1.5 maintains the original dataset's training, validation, and test set divisions, ensuring consistency in experimental comparisons with DOTA-v1.0.

DIOR-R: The DIOR-R is a specialized dataset for object detection in optical remote sensing images, consisting of 23463 images and 190288 annotated instances across 20 categories. The 20 common categories consist of airplane (APL), airport (APO), baseball field (BF), basketball court (BC), bridge (BR), chimney (CH), dam (DAM), expressway service area (ESA), expressway toll station (ETS), golf field (GF), ground track field (GTF), harbor (HA), overpass (OP), ship (SH), stadium (STA), storage tank (STO), tennis court (TC), train station (TS), vehicle (VE), and windmill(WM). Each image is standardized to 800×800 pixels with spatial resolutions ranging from 0.5 to 30 meters, and the dataset incorporates diverse imaging conditions (e.g., seasonal variations, weather effects) to enhance generalization. DIOR-R is split into training/validation (11725 images) and test (11738 images) subsets, serving as a robust benchmark for advancing object detection techniques in optical remote sensing.

B. Experimental Setting

All experiments were conducted on a computational platform equipped with an Intel Core i9-13900K CPU and a single NVIDIA GeForce RTX 3090 GPU with 24GB of memory. Experimental parameters and configuration details across different datasets are summarized in Table I. Our framework builds upon MMRotate [45], an open-source PyTorch-based toolbox tailored for oriented object detection tasks. For performance evaluation, we employ the mean Average Precision at IoU=0.5 (mAP50) metric following the PASCAL VOC 2007 protocol to assess overall detector efficacy, while per-class accuracy is validated using Average Precision (AP) scores.

In terms of training settings, we set the maximum number of iterations T to 500, considering the convergence requirements for complex and dense scenes in remote sensing imagery. Moreover, we adopt an early stopping mechanism with a tolerance of 10^{-7}. Consequently, the algorithm converges earlier in sparse scenes, thus avoiding additional computational overhead.

In addition, we conducted experimental investigations into the hyperparameters involved in the τ_s and τ_b formulas of the SGPA and SGVP methods. Fig. 3 presents the experimental results for the τ_s and τ_b hyperparameters, respectively. The findings indicate that the model achieves optimal performance when the τ_s is set to 0.65 and the τ_b is set to 0.70. Notably, when the τ_b is set to 0.10, the model encounters a matrix decomposition error, rendering the training process infeasible. Thus, this case is recorded as 0 in the relevant precision

TABLE I
EXPERIMENTAL PARAMETER SETTINGS ON FOUR DATASETS

Settings	DOTA-v1.0	DOTA-v1.5	DIOR-R	HRSC
Optimizer	AdamW	AdamW	AdamW	AdamW
Learning rate	$5e^{-5}$	$5e^{-5}$	$5e^{-5}$	$5e^{-5}$
Weight decay	0.05	0.05	0.05	0.05
Epoch decay	(8,11)	(8,11)	(24,33)	(48,66)
Epochs	12	12	36	72
Batch size	2	2	4	4
Image size	1024× 1024	1024× 1024	800× 800	800× 800

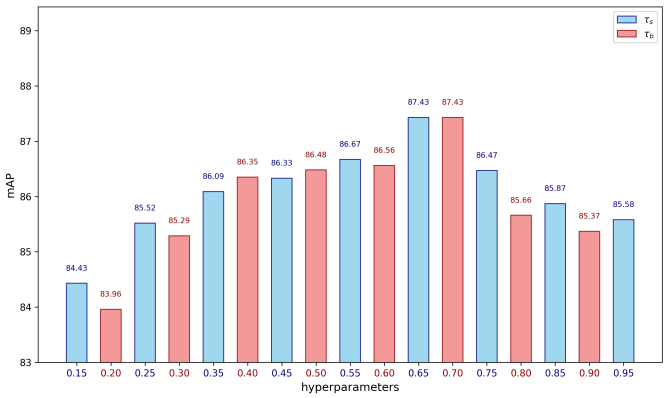


Fig. 3. The experimental study of the hyperparameter τ_s and τ_b in SGPA and SGVP on the HRSC dataset.

data. By setting a threshold of 0.65, the SGPA method can effectively filter the semantic information required for the target, thereby enhancing the perception ability of target angles and scales. On the other hand, under a threshold of 0.7, the SGVP method can select boundaries with significant gradient changes as segmentation boundaries for the watershed algorithm, while effectively suppressing weak gradient responses inside the target to reduce interference. To further verify the generalization of hyperparameters τ_s and τ_b, we conduct ablation experiments on the DIOR-R dataset in addition to the HRSC dataset. As illustrated in Fig. 4, the optimal values remain $\tau_s = 0.65$ and $\tau_b = 0.70$ on DIOR-R, consistent with the best settings obtained on HRSC. This observation demonstrates that the two hyperparameters possess good cross-dataset generalization and do not require dataset-specific fine-tuning.

C. Ablation Study

To validate the effectiveness of the proposed method, we conduct ablation studies on DIOR-R for SGPA, SGVP, and MS-Attn-MoE modules.

1) Effectiveness of SGPA: In the experiment, SGPA effectively avoids cold-start and confirmation bias during early training by leveraging multiple collaborative mechanisms. Specifically, it filters valid pixels for semantic instance assignment through confidence thresholds, while simultaneously suppressing random noise via foreground screening. Additionally, annotated-point features are employed as anchors to guide the feature clustering procedure. Beyond these operations, multi-view consistency and Gaussian-Voronoi layout constraints deliver stable self-supervised signals, which are crucial for

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

TABLE II
ABLATIVE EXPERIMENTS AND EVALUATION OF THE PROPOSED S^{2}GPD METHOD ON THE DIOR-R DATASET

Methods	SGPA	SGVP	MS-Attn-MoE	APL	APO	BF	BC	BR	CH	ESA	ETS	DAM	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
Baseline				57.3	5.0	67.1	80.8	15.3	65.0	17.0	37.4	8.7	11.2	48.5	1.4	33.4	62.6	**48.6**	57.1	80.6	18.0	22.3	12.3	37.48
	✓			57.1	**11.7**	67.7	81.2	**16.1**	67.9	26.2	**40.2**	12.9	11.2	48.6	1.7	**36.0**	55.1	35.9	57.6	80.9	21.2	**23.0**	16.6	38.43
		✓		58.6	2.9	67.7	81.2	15.2	70.0	26.7	37.7	8.3	10.4	48.8	1.7	33.5	61.5	40.1	58.7	80.6	16.7	23.0	16.5	37.99
			✓	59.3	3.3	**68.0**	81.2	12.1	71.8	**35.4**	32.1	12.7	**15.4**	34.7	9.1	24.5	60.9	47.8	58.2	80.8	**23.9**	20.2	3.2	37.73
	✓	✓		63.1	10.6	66.8	81.3	12.1	70.8	23.5	38.9	13.9	13.8	45.4	1.5	27.5	**64.5**	39.4	**64.4**	81.3	20.3	21.0	**17.5**	38.88
		✓	✓	57.9	7.0	66.7	**81.3**	12.5	**71.8**	27.8	38.7	14.2	10.6	47.5	**9.1**	29.2	61.9	41.2	61.2	81.0	17.3	21.4	13.1	38.56
	✓		✓	62.1	11.5	65.4	81.1	12.7	70.9	23.5	39.1	14.1	11.0	**50.4**	1.6	30.0	64.1	39.5	62.9	81.2	21.0	21.3	16.9	39.01
	✓	✓	✓	**63.1**	10.8	66.7	81.2	12.2	71.0	25.6	39.4	**14.2**	14.4	48.9	9.1	29.1	64.4	40.0	64.0	**81.3**	21.5	20.9	17.3	**39.75**

TABLE III
COMPARISON WITH STATE-OF-THE-ART POINT-SUPERVISED OOD METHODS ON THE DOTA-V1.0 DATASET

Methods	*	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
Point2Mask-RBox [46]	×	4.0	23.1	3.8	1.3	15.1	1.0	3.3	19.0	1.0	29.1	0.0	9.5	7.4	21.1	7.1	9.72
P2BNet+H2RBox [47]	×	24.7	35.9	7.1	27.9	3.3	12.1	17.5	17.5	0.8	34.0	6.3	49.6	11.6	27.2	18.8	19.63
P2BNet+H2RBox-v2 [47]	×	11.0	44.8	14.9	15.4	36.8	16.7	27.8	12.1	1.8	31.2	3.4	50.6	12.6	36.7	12.5	21.87
P2RBox (FCOSR) [38]	×	87.8	65.7	15.0	**60.7**	73.0	71.7	78.9	81.5	44.5	81.2	**41.2**	39.3	45.5	57.5	41.2	59.04
PointOBB (FCOSR) [6]	×	26.1	65.7	9.1	59.4	65.8	34.9	29.8	0.5	2.3	16.7	0.6	49.0	21.8	41.0	36.7	30.08
Point2RBox [10]	✓	62.9	64.3	14.4	35.0	28.2	38.9	33.3	25.2	2.2	44.5	3.4	48.1	25.9	45.0	22.6	34.07
Point2RBox+SK [10]	✓	53.3	63.9	3.7	50.9	40.0	39.2	45.7	76.7	10.5	56.1	5.4	49.5	24.2	51.2	33.8	40.27
Point2RBox+SK [10]	×	66.4	59.5	5.2	52.6	54.1	53.9	57.3	**90.8**	3.2	57.8	6.1	47.4	22.9	55.7	40.5	44.90
PointOBB-v2 (FCOSR) [11]	×	64.5	27.8	1.9	36.2	58.8	47.2	53.4	90.5	62.2	45.3	12.1	41.7	8.1	43.7	32.0	41.68
PointOBB-v3 [12]	✓	30.9	39.4	13.5	22.7	61.2	7.0	43.1	62.4	59.8	47.3	2.7	45.1	16.8	55.2	11.4	41.29
PointOBB-v3 (FCOSR) [12]	×	52.9	54.4	21.3	52.7	65.6	44.9	67.8	87.2	26.7	73.4	32.6	**53.3**	39.0	56.4	10.2	49.24
Point2RBox-v2 [13]	✓	78.4	52.7	8.3	40.9	71.0	60.5	74.7	88.7	65.5	72.1	24.4	26.1	30.1	50.7	21.0	51.00
Point2RBox-v2 (FCOSR) [13]	×	88.0	**72.6**	8.0	46.2	**79.6**	**76.3**	86.9	89.1	**79.7**	82.9	26.2	45.3	45.8	**66.3**	**46.3**	62.61
S^{2}GPD	✓	78.2	51.0	20.2	41.5	71.5	64.8	74.4	88.1	62.2	73.9	15.4	31.5	30.3	51.6	26.9	52.12
S^{2}GPD (FCOSR)	×	**88.0**	71.9	**27.9**	48.4	79.1	75.9	**87.4**	89.8	70.4	**83.3**	21.0	47.2	**45.8**	65.6	45.4	**63.14**

* denotes training method. ✓is an end-to-end training approach. × is the training paradigm for pseudo-labels.

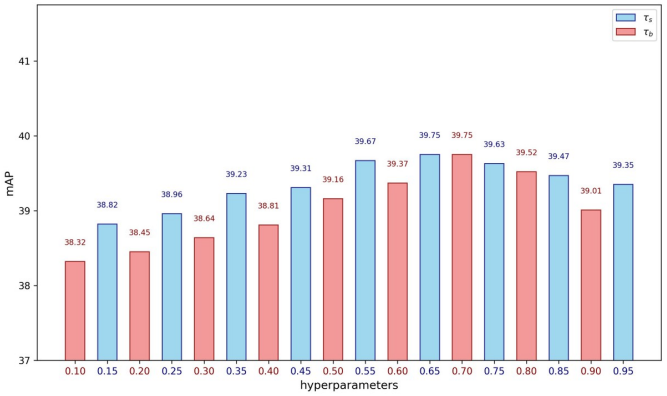


Fig. 4. The experimental study of the hyperparameters τ_s and τ_b in SGPA and SGVP on the DIOR-R dataset.

learning object scale and angle. Furthermore, ground-truth supervision for object centers serves as an extra safeguard, actively preventing bias accumulation throughout training. Given these design merits, we evaluate SGPA's practical performance on the DIOR-R dataset. As shown in Table II, the standalone application of the semantic-guided point assignment strategy improves precision by 0.95%, with notable enhancements observed for large-scale targets such as BF, APO, CH, ESA, ETS, DAM, OP, TS, and WM. These results indicate that SGPA has an effect on enhancing semantic supplementation for large-sized targets, effectively improving the system's perception ability for large-scale targets. Furthermore, the combination of SGPA with SGVP further optimizes the perception ability for small targets. By leveraging the structural prior information provided by SGVP, SGPA enhances the recognition accuracy of target contours.

2) Effectiveness of SGVP: As revealed in Table II, the experiment independently validates the effectiveness of SGVP. Compared to the baseline model, SGVP achieves a 0.51% performance improvement, with enhancements particularly observed in categories such as APL, CH, ESA, and STO. SGVP enhances the recognition of contour boundaries by performing semantic filling on the boundaries of larger targets while suppressing weak responses at internal boundaries. Furthermore, the combination of SGVP with MS-Attn-MoE yields a 0.57% precision improvement compared to standalone SGVP. This enhancement is primarily reflected in categories including APO, DAM, and STO, indicating that the integration of MS-Attn-MoE further strengthens the model's performance.

3) Effectiveness of MS-Attn-MoE: As revealed in Table II, compared to the baseline model, MS-Attn-MoE achieves performance improvements across multiple categories, including APL, BF, CH, ESA, DAM, HA, and TS. These categories encompass targets of varying sizes, ranging from large to small-scale objects. This outcome underscores that MS-Attn-MoE primarily benefits from its multi-scale feature extraction capability, which enhances the precise perception of diverse target sizes. In addition, the fusion of MS-Attn-MoE with SGPA yields a substantial 0.58% performance improvement, with gains observed in categories such as APL, CH, DAM, SH, STA, and STO.This enhancement is mainly attributed to the integration of classification heatmap semantic information, which further optimizes the performance of the point-supervised model by complementing the multi-scale target perception framework. The synergistic effect of multi-scale feature learning and semantic-guided point assignment thus collectively contributes to the superior performance of the fused model.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

TABLE IV
COMPARISON WITH STATE-OF-THE-ART POINT-SUPERVISED OOD METHODS ON THE DIOR-R DATASET

Methods	*	APL	APO	BF	BC	BR	CH	ESA	ETS	DAM	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
Point2Mask-RBox [46]	×	15.6	0.1	50.6	25.4	4.2	50.9	23.8	17.5	8.1	0.8	9.6	1.4	15.3	1.6	5.8	6.3	17.9	7.1	4.5	9.2	13.77
P2BNet+H2RBox [47]	×	52.7	0.1	60.6	80.0	0.1	22.6	11.5	5.2	0.7	0.2	42.8	2.8	0.2	25.1	8.6	29.1	69.8	9.6	7.4	22.6	22.59
P2BNet+H2RBox-v2 [47]	×	51.6	3.0	65.2	78.3	0.1	8.1	7.6	6.3	0.8	0.3	44.9	2.3	0.1	35.9	9.3	39.2	79.0	8.8	10.3	21.3	23.61
Point2RBox [10]	✓	42.0	9.1	62.9	52.8	10.8	72.2	3.0	43.9	5.5	9.7	25.1	9.1	21.0	24.0	20.4	25.1	71.7	4.5	16.1	16.3	27.34
PointOBB (FCOSR) [6]	×	58.4	**17.1**	**70.7**	77.7	0.1	70.3	64.7	4.5	7.2	0.8	74.2	9.9	9.1	69.0	38.2	49.8	46.1	16.8	32.4	29.6	37.31
PointOBB (ORCNN) [6]	×	58.2	15.3	70.5	78.6	0.1	72.2	**69.6**	1.8	3.7	0.3	77.3	**16.7**	4.0	79.2	39.6	51.7	44.9	16.8	**33.6**	27.7	38.06
PointOBB-v2 (FCOSR) [11]	×	45.8	6.4	58.2	66.5	2.4	60.9	40.3	33.2	4.5	55.3	34.2	11.4	8.9	43.0	46.8	37.3	76.9	21.9	10.6	19.8	34.35
PointOBB-v2 (ORCNN) [11]	×	41.5	6.0	59.7	79.7	6.7	72.2	50.0	41.0	5.8	**59.4**	38.3	11.2	21.6	53.0	54.4	49.3	80.9	**33.2**	11.1	17.2	39.62
PointOBB-v3 [12]	✓	53.1	11.0	70.4	79.0	0.1	70.1	53.0	1.6	0.9	45.7	69.4	3.1	3.4	63.2	44.5	57.3	58.1	1.3	30.3	36.5	37.60
PointOBB-v3 (FCOSR) [12]	×	37.1	13.0	70.2	79.4	0.1	71.5	44.7	50.8	4.5	52.9	74.7	2.5	0.5	69.6	46.1	58.0	58.9	3.0	31.8	34.4	40.18
PointOBB-v3 (ORCNN) [12]	×	32.8	11.2	69.9	80.3	0.1	**72.3**	42.1	**55.3**	0.7	54.8	**77.9**	1.9	0.8	79.8	53.3	60.5	68.1	0.9	33.2	**40.5**	41.82
Point2RBox-v2 [13]	✓	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	34.70
Point2RBox-v2 (FCOSR) [13]	×	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	44.45
S^2GPD	✓	63.1	10.8	66.7	81.2	12.2	71.0	25.6	39.4	14.2	14.4	48.9	9.1	29.1	64.4	40.0	64.0	**81.3**	21.5	20.9	17.3	39.75
S^2GPD (FCOSR)	×	**69.0**	7.3	68.7	**81.3**	**13.9**	71.9	43.5	42.4	**17.0**	17.5	56.3	9.1	**35.9**	**79.8**	**69.6**	**69.2**	80.8	29.2	27.8	13.8	**45.20**

* denotes training method. ✓is an end-to-end training approach. × is the training paradigm for pseudo-labels.

D. Comparison With the State-of-the-Art

To demonstrate the superior performance of the S^2GPD method, this experiment conducts extensive comparative experiments on the DOTA-v1.0, DOTA-v1.5, DIOR-R, and HRSC datasets. All experimental parameters of the comparative methods follow the default settings provided in relevant academic papers. In the process of pseudo-label training using labels generated by the inference of the FCOSR model and the end-to-end model, this process is abbreviated as (FCOSR). Meanwhile, the (ORCNN) denotes using the ORCNN model for pseudo-label training.

1) Results on DOTA-v1.0: Table III provides a detailed comparison of the performance of various point-supervised methods, with a comparative analysis against models incorporating manual prior knowledge. The observations show that S^2GPD achieves a 1.12% performance improvement over Point2RBox-v2, which is primarily reflected in key categories such as BR, LV, RA, and HC. In particular, the BR category exhibits an improvement of up to 11.9%. Although this gain is appreciable, it mainly reflects the effectiveness of the structural prior and watershed mechanism in partially compensating for the representation of irregular objects. However, a single anisotropic Gaussian may not fully capture complex contours such as curvature and concavity for non-convex targets like bridges and harbors. Furthermore, S^2GPD trained with pseudo-labels (FCOSR) even outperforms the P2RBox model integrated with SAM. This further demonstrates that excellent results can be achieved by relying solely on the available information inherent in images without introducing any prior knowledge.

2) Results on DIOR-R: As revealed in Table IV, this experiment conducts a comparative analysis of the proposed model and existing point-supervised models on the DIOR-R dataset, focusing on their detection performance across diverse object categories. The end-to-end S^2GPD method demonstrates significant superiority over Point2RBox-v2, particularly in scenarios requiring precise localization of complex-shaped targets. However, this result still falls below the performance of S^2GPD (FCOSR). Furthermore, S^2GPD trained with pseudo-labels (FCOSR) exhibits competitive performance even against some ORCNN-based pseudo-labeled models, especially in categories with distinct visual patterns. The consistent outperformance of S^2GPD across both end-to-end and pseudo-labeled training paradigms highlights its robustness in leveraging point supervision signals, even when compared to models that rely on more complex architectural modifications or external prior knowledge. As shown in Fig. 5, we compare the detection performance of the S^2GPD against Point2RBox and Point2RBox-v2 on the DIOR-R dataset. Experimental results demonstrate that S^2GPD, leveraging a semantic guidance strategy, achieves superior precision in target scale and angle estimation under dense distributions and sparse scenarios for certain categories.

TABLE V
COMPARISON WITH STATE-OF-THE-ART POINT-SUPERVISED OOD METHODS ON HRSC AND DOTA-V1.5 DATASETS.

Methods	*	HRSC	DOTA-v1.5
PointOBB (FCOSR) [6]	×	-	10.66
PointOBB (ORCNN) [6]	×	-	10.92
Point2RBox+SK [10]	✓	79.40	30.51
PointOBB-v2 (FCOSR) [11]	×	-	30.59
Point2RBox-v2 [13]	✓	82.67	39.45
PointOBB-v2 (ORCNN) [11]	×	-	32.01
Wholly-WOOD [48]	×	87.30	52.78
PointOBB-v3 [12]	✓	-	31.25
PointOBB-v3 (FCOSR) [12]	×	-	33.79
Point2RBox-v2 (FCOSR) [13]	×	86.15	54.06
PointOBB-v3 (ORCNN) [12]	×	-	38.08
S^2GPD	✓	87.43	43.25
S^2GPD (FCOSR)	×	**88.10**	**54.52**

* denotes training method. ✓is an end-to-end training approach. × is the training paradigm for pseudo-labels.

3) Results on Other Datasets: As illustrated in Table V, we further validate the generalization capability of the S^2GPD model on two authoritative datasets, DOTA-v1.5 and HRSC. Compared to the end-to-end trained Point2RBox-v2 baseline, the proposed method achieves a 3.8% accuracy improvement on DOTA-v1.5 and a more significant 4.76% gain on HRSC, demonstrating its superior performance across different scenarios. Notably, on the HRSC dataset, the S^2GPD outperforms the pseudo-label trained results of Point2RBox-v2 (FCOSR) through end-to-end training alone, highlighting its efficient feature learning capability without relying on external label refinement. These results not only confirm the advancement of S^2GPD but also underscore its potential for advancing object detection tasks in complex visual environments. As demonstrated in Fig. 6, the S^2GPD improves detection performance by reducing false positives and omissions compared to Point2RBox-v2. As shown in Fig.7, this structure guides the Voronoi partitioning method on DOTA-v1.5 dataset. First, the object centers are used as seed points to generate initial Voronoi cells based on an anisotropic Gaussian probability

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

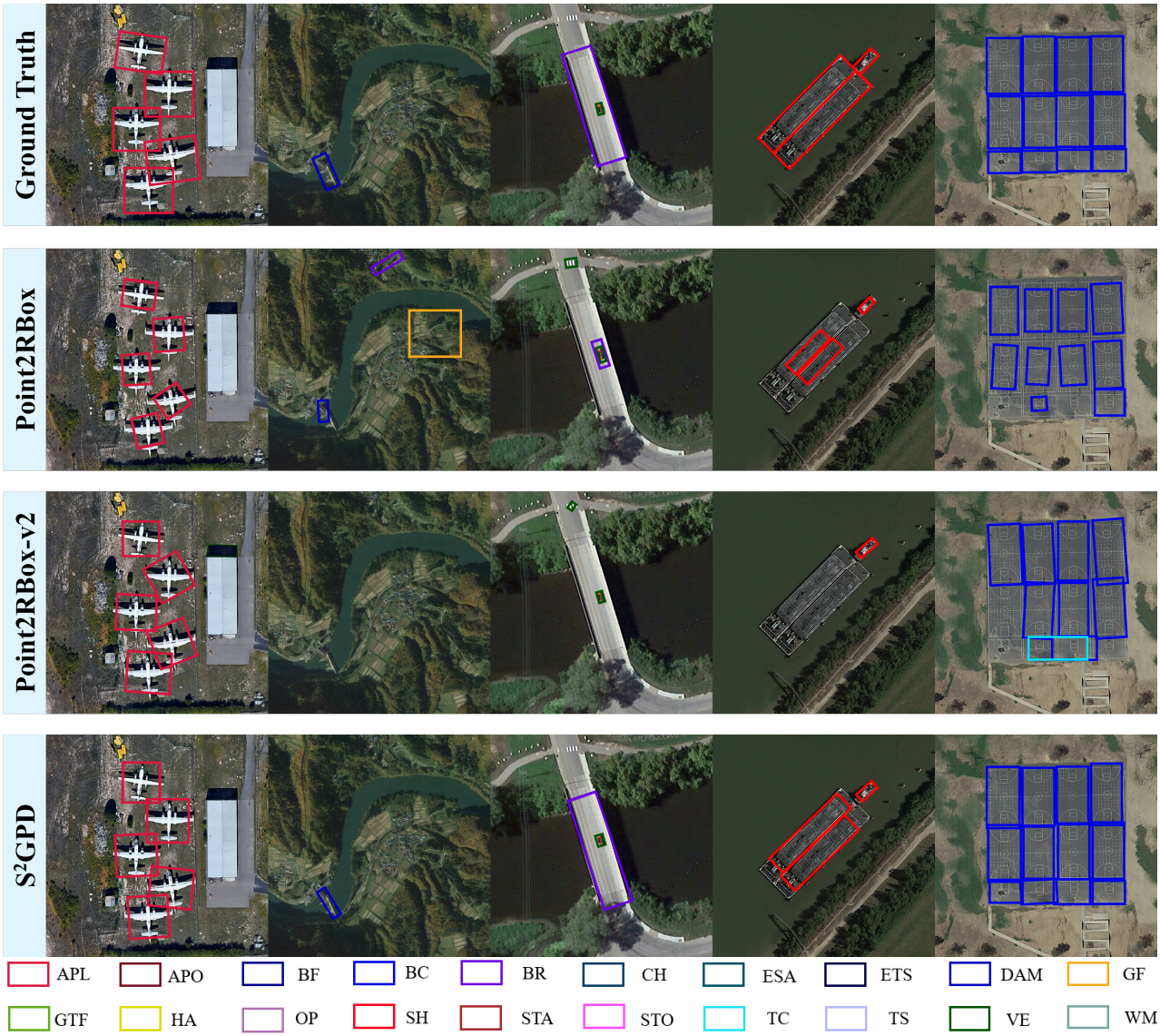


Fig. 5. The comparison of Point2RBox, Point2RBox-v2, and S^{2}GPD methods on the DIOR-R dataset. These methods all employ end-to-end training.

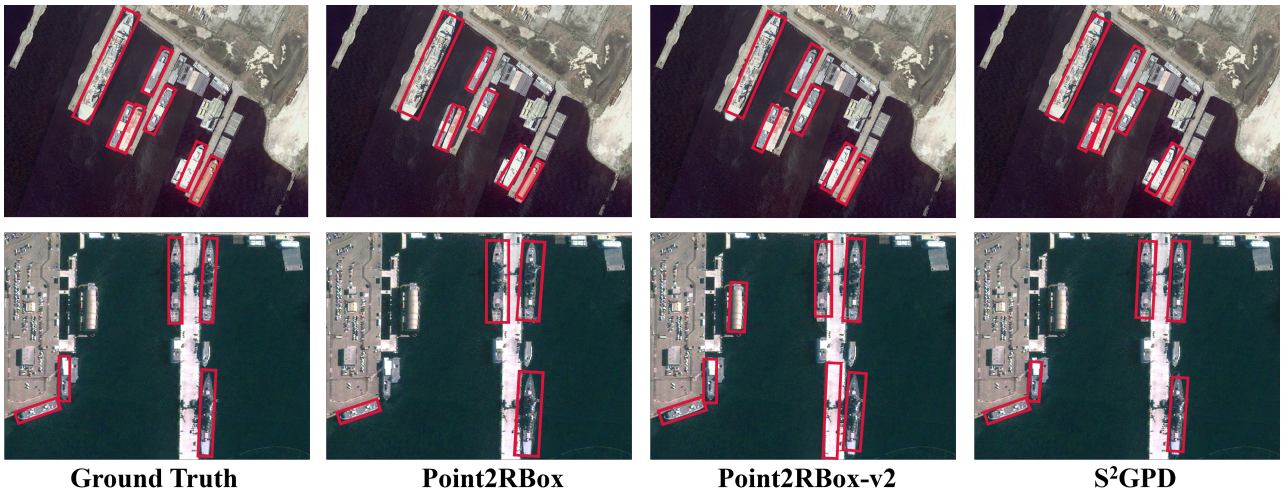
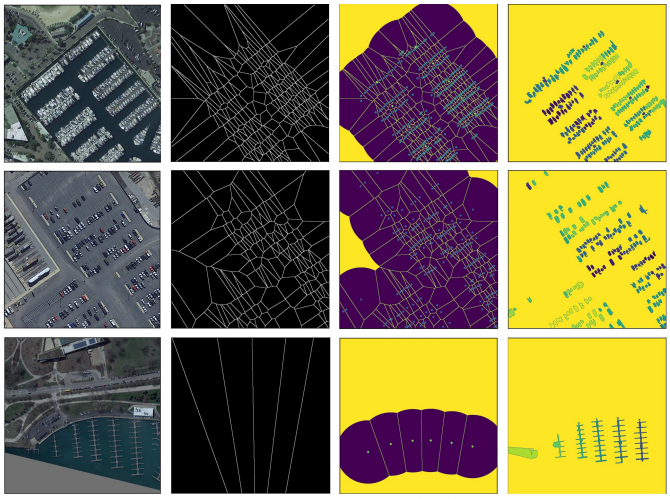


Fig. 6. The comparison of Point2RBox, Point2RBox-v2, and S^{2}GPD methods on the HRSC dataset. These methods all employ end-to-end training.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325



(a) Input image (b) Ridge map (c) Voronoi partition (d) Watershed segmentation

Fig. 7. Visualization of Voronoi partition process.

distribution. The gradient structural prior from the classification heatmap is then fused to refine the boundary ridges. Subsequently, the image is divided into definite foreground, background, and transitional regions according to a probability threshold. Using these marker regions as constraints, the watershed algorithm yields refined instance segmentation results. Finally, precise object contours are obtained, providing shape and scale supervision for the detection model.

V. Conclusion

In this study, we propose a single-point supervised oriented object detection model that exploits classification semantic and structural prior information to alleviate scale and angle estimation errors under complex scenarios. Specifically, the SGPA strategy adopts class-aware semantic heatmaps to guide instance-level clustering and assign high-value prior points, compensating for missing object-level semantics and improving scale and angle prediction accuracy. The MS-Attn-MoE module enhances perception of scale-variant objects via multi-scale convolutions, and adaptively fuses multi-scale features onto one feature map for more accurate scale prediction in point-supervised settings. The SGVP integrates structural priors into Voronoi partitioning for better target shape perception, precise contour segmentation, and strengthened network learning. It is worth noting that the proposed method has several main limitations. First, the performance of SGVP partially depends on properly set hyperparameter thresholds for separating foreground and background regions. Therefore, designing adaptive thresholds to replace manually specified hyperparameters constitutes a promising direction for further investigation. In addition, training memory consumption rises when the number of objects grows dramatically or the data exhibits extremely dense object distributions. Hence, another focus of future work is to explore more efficient algorithms to enable the framework for edge deployment.

References

[1] K. Ye, H. Tang, B. Liu *et al.*, "Codrone: A comprehensive oriented object detection benchmark for uav," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1-15, 2026.

[2] T. Niu, Q. Teng, X. He *et al.*, "Phenotype-aware imitation learning for small, oriented object detection in remote sensing," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1-21, 2026.

[3] Z. Ding, Y. Zhou, J. Zhao *et al.*, "Real-time oriented object detection transformer in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1-14, 2026.

[4] J. Lu, Q. Hu, R. Zhu *et al.*, "AFWS: Angle-free weakly supervised rotating object detection for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1-14, 2024.

[5] C. Fu, J. Chen, and Y. Zhu, "Line2Rbox: Line-supervised oriented object detection," 2025. [Online]. Available: https://openreview.net/forum?id=ivLD3sCDcG

[6] J. Luo, X. Yang, Y. Yu *et al.*, "PointOBB: Learning oriented object detection via single point supervision," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 16730-16740.

[7] L. Chen, T. Yang, X. Zhang *et al.*, "Points as Queries: Weakly semi-supervised object detection by points," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 8819-8828.

[8] S. Zhang, J. Long, Y. Xu *et al.*, "PMHO: Point-supervised oriented object detection based on segmentation-driven proposal generation," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1-18, 2024.

[9] A. Kirillov, E. Mintun, N. Ravi *et al.*, "Segment anything," *arXiv preprint arXiv:2304.02643*, 2023.

[10] Y. Yu, X. Yang, Q. Li *et al.*, "Point2RBox: Combine knowledge from synthetic visual patterns for end-to-end oriented object detection with single point supervision," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 16783-16793.

[11] B. Ren, X. Yang, Y. Yu *et al.*, "PointOBB-v2: Towards simpler, faster, and stronger single point supervised oriented object detection," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.

[12] P. Zhang, J. Luo, X. Yang *et al.*, "PointOBB-v3: Expanding performance boundaries of single point-supervised oriented object detection," *Int. J. Comput. Vision*, vol. 133, no. 9, pp. 6108-6128, May 2025.

[13] Y. Yu, B. Ren, P. Zhang *et al.*, "Point2RBox-v2: Rethinking point-supervised oriented object detection with spatial layout among instances," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 19283-19293.

[14] J. Han, J. Ding, N. Xue *et al.*, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 2785-2794.

[15] Y. Wang, Z. Zhang, W. Xu *et al.*, "Learning oriented object detection via naive geometric computing," *IEEE Trans. Neural Netw. Learn. Syst.*, pp. 1-13, 2023.

[16] Y. Han, W. Zhan, D. Huo *et al.*, "TSFDet: Task-specific feature disentanglement for two-stage oriented object detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1-14, 2026.

[17] T.-Y. Lin, P. Goyal, R. Girshick *et al.*, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2017, pp. 2999-3007.

[18] J. Xie, Y. Pang, J. Nie *et al.*, "Latent feature pyramid network for object detection," *IEEE Trans. Multimedia*, vol. 25, pp. 2153-2163, 2023.

[19] Y. Du, L. Du, Y. Guo *et al.*, "Adaptive anchor-based detector with constrained rirconv for oriented vehicles in sar images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1-17, 2025.

[20] C. Fu and Y. Zhu, "Oaadet: One-stage anchor-free arbitrary-oriented object detector via center-ness shift correction," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1-16, 2026.

[21] P. Dai, S. Yao, Z. Li *et al.*, "ACE: Anchor-free corner evolution for real-time arbitrarily-oriented object detection," *IEEE Trans. Image Process.*, vol. 31, pp. 4076-4089, 2022.

[22] Z. Li, W. Liu, Z. Xie *et al.*, "FAA-Det: Feature augmentation and alignment for anchor-free oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1-11, 2024.

[23] Y. Zeng, Y. Chen, X. Yang *et al.*, "ARS-DETR: Aspect ratio-sensitive detection transformer for aerial oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, pp. 1-15, 2024.

[24] G. Cao, W. Huang, X. Lan *et al.*, "Cross-dino: Cross the deep mlp and transformer for small object detection," *IEEE Trans. Multimedia*, vol. 27, pp. 7369-7379, 2025.

[25] Y. Hu, B. Hou, C. Yang *et al.*, "FCG-DETR: Foreground confidence-guided detection transformer for remote sensing oriented object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 64, pp. 1-17, 2026.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

This article has been accepted for publication in IEEE Transactions on Geoscience and Remote Sensing. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TGRS.2026.3750325

[26] J. Ding, N. Xue, Y. Long *et al.*, "Learning roi transformer for oriented object detection in aerial images," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2019, pp. 2844-2853.
[27] X. Xie, G. Cheng, J. Wang *et al.*, "Oriented r-cnn for object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2021, pp. 3500-3509.
[28] W. Qian, X. Yang, S. Peng *et al.*, "RSDet++: Point-based modulated loss for more accurate rotated object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 32, no. 11, pp. 7869-7879, 2022.
[29] X. Yang, J. Yan, Z. Feng *et al.*, "R^3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, vol. 35, 2021, pp. 3163-3171.
[30] Y. Yao, G. Cheng, C. Lang *et al.*, "Hierarchical mask prompting and robust integrated regression for oriented object detection," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 34, no. 12, pp. 13071-13084, 2024.
[31] W. Li, Y. Chen, K. Hu *et al.*, "Oriented reppoints for aerial object detection," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 1819-1828.
[32] Y. Liang, J. Feng, X. Zhang *et al.*, "MidNet: An anchor-and-angle-free detector for oriented ship detection in aerial images," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 61, pp. 1-13, 2023, Art no. 5612113.
[33] Y. Zhu, Y. Ruan, L. Yang *et al.*, "RotaTR: Detection transformer for dense and rotated object," *arXiv preprint arXiv:2312.02821*, 2023.
[34] H. Lee, M. Song, J. Koo *et al.*, "Hausdorff distance matching with adaptive query denoising for rotated detection transforme," *arXiv preprint arXiv:2305.07598*, 2023.
[35] X. Yang, G. Zhang, W. Li *et al.*, "H2RBox: Horizontal box annotation is all you need for oriented object detection," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2023.
[36] Y. Yu, X. Yang, Q. Li *et al.*, "H2RBox-v2: Incorporating symmetry for boosting horizontal box supervised oriented object detection," in *Proc. Adv. Neural Inf. Process. Syst. (NeuralIPS)*, vol. 36, 2023, pp. 59137-59150.
[37] L. Wang, Y. Zhan, X. Lin, B. Yu, L. Ding, J. Zhu, and D. Tao, "Explicit and implicit box equivariance learning for weakly-supervised rotated object detection," *IEEE Trans. Emerging Top. Comput. Intell.*, vol. 9, no. 1, pp. 509-521, 2025.
[38] G. Cao, X. Yu, W. Yu *et al.*, "P2RBox: Point prompt oriented object detection with sam," *arXiv preprint arXiv:2311.13128v2*, 2023.
[39] J. Feng, Y. Liang, X. Zhang *et al.*, "SDANet: Semantic-embedded density adaptive network for moving vehicle detection in satellite videos," *IEEE Transactions on Image Processing*, vol. 32, pp. 1788-1801, 2023.
[40] J. Feng, Q. Jiang, J. Zhang *et al.*, "CFDRM: Coarse-to-fine dynamic refinement model for weakly supervised moving vehicle detection in satellite videos," *IEEE Transactions on Geoscience and Remote Sensing*, vol. 62, pp. 1-13, 2024, Art no. 5626413.
[41] X. Yang, J. Yan, Q. Ming *et al.*, "Rethinking rotated object detection with gaussian wasserstein distance loss," in *Proc. Int. Conf. Mach. Learn. (ICML)*, 2021, pp. 11830-11841.
[42] G.-S. Xia, X. Bai, J. Ding *et al.*, "DOTA: A large-scale dataset for object detection in aerial images," in *Proc. IEEE Int. Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 3974-3983.
[43] K. Li, G. Wan, G. Cheng *et al.*, "Object detection in optical remote sensing images: A survey and a new benchmark," *arXiv preprint arXiv:1909.00133*, 2019.
[44] Z. Liu, L. Yuan, L. Weng, and Y. Yang, "A high resolution optical satellite image dataset for ship recognition and some new baselines," in *Proc. Int. Conf. Pattern Recognit. Appl. Methods. (ICPRAM)*, Jan. 2017, pp. 324-331.
[45] Y. Zhou, X. Yang, G. Zhang *et al.*, "MMRotate: A rotated object detection benchmark using pytorch," in *Proc. ACM Int. Conf. Multimedia.*, 2022, pp. 7331-7334.
[46] W. Li, Y. Yuan, S. Wang *et al.*, "Point2Mask: Point-supervised panoptic segmentation via optimal transport," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 572-581.
[47] P. Chen, X. Yu, X. Han *et al.*, "P2Object: Single point supervised object detection and instance segmentation," *Int. J. Comput. Vision.*, vol. 133, no. 8, pp. 5544-5568, 2025.
[48] Y. Yu, X. Yang, Y. Li *et al.*, "Wholly-WOOD: Wholly leveraging diversified-quality labels for weakly-supervised oriented object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 6, pp. 4438-4454, 2025.

Yifan Xi received the M.S. degree in information and communication engineering from Chang'an University, Xi'an, China, in 2021. He is currently pursuing the Ph.D. degree in control science and engineering with Hunan University, Changsha, China. His research interests focus on computer vision and pattern recognition, especially object detection.

Ting Lu (Member, IEEE) received the B.S. and Ph.D. degrees from Hunan University, Changsha, China, in 2011 and 2017, respectively.

From 2014 to 2015, she was a Visiting Ph.D. Student with the Department of Information Engineering and Computer Science, University of Trento, Trento, Italy, supported by the China Scholarship Council. In 2017, she became an Assistant Professor with Hunan University, where she is currently an Associate Professor with the College of Electrical and Information Engineering. Her research interests include image processing, pattern recognition, and applications in remote sensing.

Shutao Li (Fellow, IEEE) received the B.S., M.S., and Ph.D. degrees from Hunan University, Changsha, China, in 1995, 1997, and 2001, respectively, all in electrical engineering.

In 2001, he joined the College of Electrical and Information Engineering, Hunan University. From May 2001 to October 2001, he was a Research Associate with the Department of Computer Science, The Hong Kong University of Science and Technology, Hong Kong. From November 2002 to November 2003, he was a Post-Doctoral Fellow with the Royal Holloway College, University of London, London, U.K. From April 2005 to June 2005, he was a Visiting Professor with the Department of Computer Science, The Hong Kong University of Science and Technology. He is currently a Full Professor with the College of Electrical and Information Engineering, Hunan University. He has authored or coauthored over 200 refereed papers. His current research interests include image processing, pattern recognition, and artificial intelligence.

Dr. Li gained two Second-Grade State Scientific and Technological Progress Awards of China in 2004 and 2006. He is an Associate Editor of the IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and the IEEE TRANSACTIONS ON INSTRUMENTATION AND MEASUREMENT. He is a member of the Editorial Board of the Information Fusion and the Sensing and Imaging.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 09,2026 at 21:33:26 UTC from IEEE Xplore. Restrictions apply.
© 2026 IEEE. All rights reserved, including rights for text and data mining and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.